

SIVANA HAMER

Ph.D. Student in Computer Science | Researching Software Supply Chain Security

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Sivana Hamer

Fourth-year Computer Science Ph.D. Student at North Carolina State University. I am currently researching the **state of software supply chain security** as a community to **help improve the security posture of industry and open-source projects**. I have published in top software engineering venues such as ICSE, TSE, FSE, and TOSEM. I am also a Google and Goodnight PhD fellow. I look forward to opportunities to conduct software supply chain security research.

RECENT EXPERIENCE

Graduate Research Assistant 📅 Aug 2023–Present
North Carolina State University

Technical Intern 📅 May 2026–Aug 2026
Applied Research Associates

Security Research Intern 📅 May 2025–Aug 2025
Phoenix Security

FEATURED RESEARCH PROJECTS

Reduce Your Risk of Being Solarwinds, Log4j, or XZ Utils

- Analyzed the MITRE ATT&CK attack techniques in SolarWinds, Log4j, and XZ Utils to systematically synthesize software supply chain framework tasks to provide software organizations with a recommended starter kit of tasks. Collaboration with **Yahoo**.
- Methods:** Qualitative Analysis, Incident Analysis, Meta Synthesis.
- Results:** Frameworks are missing tasks; hence, even if all tasks were adopted, organizations would still be vulnerable to attacks.
- Publication:** International Conference on Software Engineering.

Reputation Measures to Review Dependencies

- Investigated if network centrality measures, proxying contributor reputation, can be used as a signal to inform developers of dependency changes that require additional examination.
- Methods:** Mixed-Methods, Statistical Models, Social Networks.
- Results:** Network centrality measures are a significant factor in explaining how developers review dependencies in Rust.
- Publication:** IEEE Transactions on Software Engineering.

Comparing Vulnerabilities ChatGPT and StackOverflow

- Compared the vulnerabilities of ChatGPT and StackOverflow to help raise software developers' awareness of the security implications when selecting code snippet platforms.
- Method:** Quantitative Analysis, Statistical Methods.
- Results:** ChatGPT generated less vulnerable code. Yet, insecure code propagation can happen in both platforms.
- Tools:** Python, R, ChatGPT API, StackOverflow API, CodeQL.
- Publication:** IEEE Security and Privacy Workshops.

RESEARCH INTERESTS

Software Measurement • Software Security
• Empirical Software Engineering • System Security

EDUCATION

Ph.D. Computer Science
North Carolina State University

📅 Aug 2023 – Present

Advisors: Dr. Laurie Williams & Dr. William Enck

M.Sc. Computer Science

Universidad de Costa Rica

📅 2023

Thesis: Mining software repositories to automatically measure developer code contributions. *Advisor:* Dr. Christian Quesada-López

B.Sc. Computer Science

Universidad de Costa Rica

📅 2020

FEATURED AWARDS

- Google PhD Fellowship (2025-2027).
- Goodnight Doctoral Fellowship (2023-2027).
- ICSE Distinguished Shadow Reviewer (2026).
- RSA Conference Security Scholar (2024).
- North Carolina State University Provost's Doctoral Fellowship (2023).

SKILLS

- Languages:** English, Spanish.
- Programming languages:** Python, Java, R, C#, JavaScript, Bash, SQL.
- Software tools:** Git, Jenkins, JIRA, Visual Studio Code, CodeQL, SonarQube, LLMs.
- Frameworks and libraries:** ASP.NET, Flask, Bootstrap, jQuery, React, Unity, n8n.
- Research methods:** Quantitative, Qualitative, Mining Software Repositories, Machine Learning, Statistical Models.